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[EOD REPORT] Anuj Kumar - 24/02/2026

1 message

Anuj Kumar <anuj.kumar@blubridge.com> 24 February 2026 at 20:10
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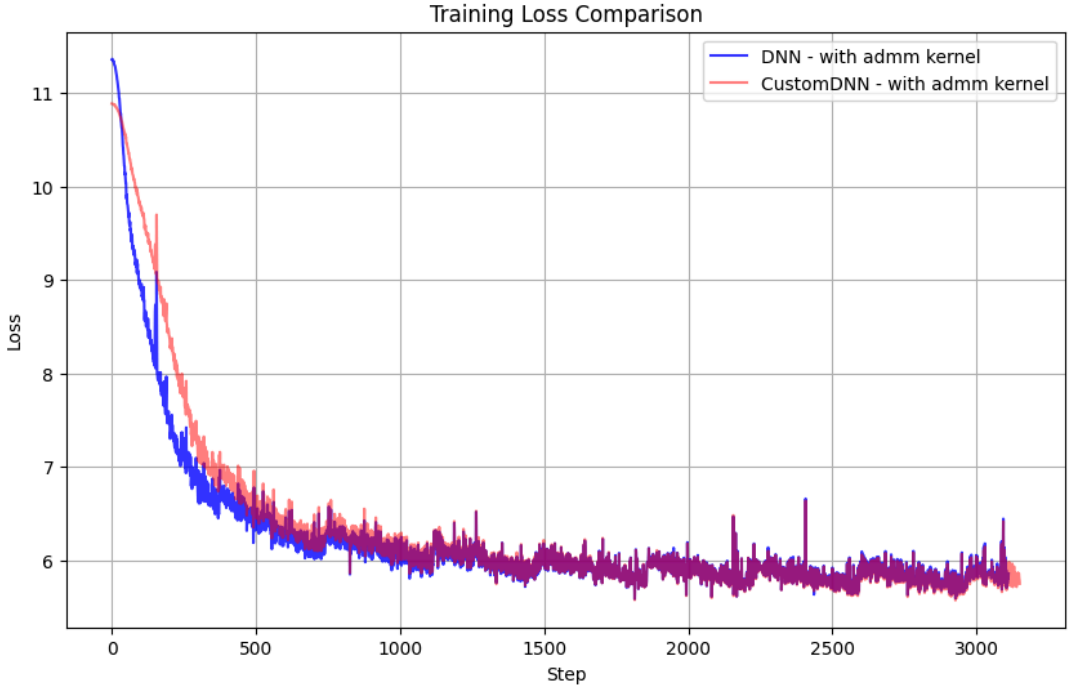
Benchmarked my module against Adhitya's module.

First used "addmm" kernel for both



The graph overlapped nicely for the 2 runs, although there was a deviation initially.

```
Maximum throughput (DNN - with admm kernel):  49777.66767
Avg. Throughput (DNN - with admm kernel): 47432.84217088271
Min Train Loss (DNN - with admm kernel): 5.587
Average loss (DNN - with admm kernel):  6.245377012210798
Max Grad Norm (DNN - with admm kernel):  3.782948
Min Val Loss (DNN - with admm kernel): 5.841319

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Maximum throughput (CustomDNN - with admm kernel):  43534.666343
Avg. Throughput (CustomDNN - with admm kernel): 42866.60100785878
Min Train Loss (CustomDNN - with admm kernel): 5.573395
Average loss (CustomDNN - with admm kernel):  6.329750897492859
Max Grad Norm (CustomDNN - with admm kernel):  2.872173
Min Val Loss (CustomDNN - with admm kernel): 5.820093
```
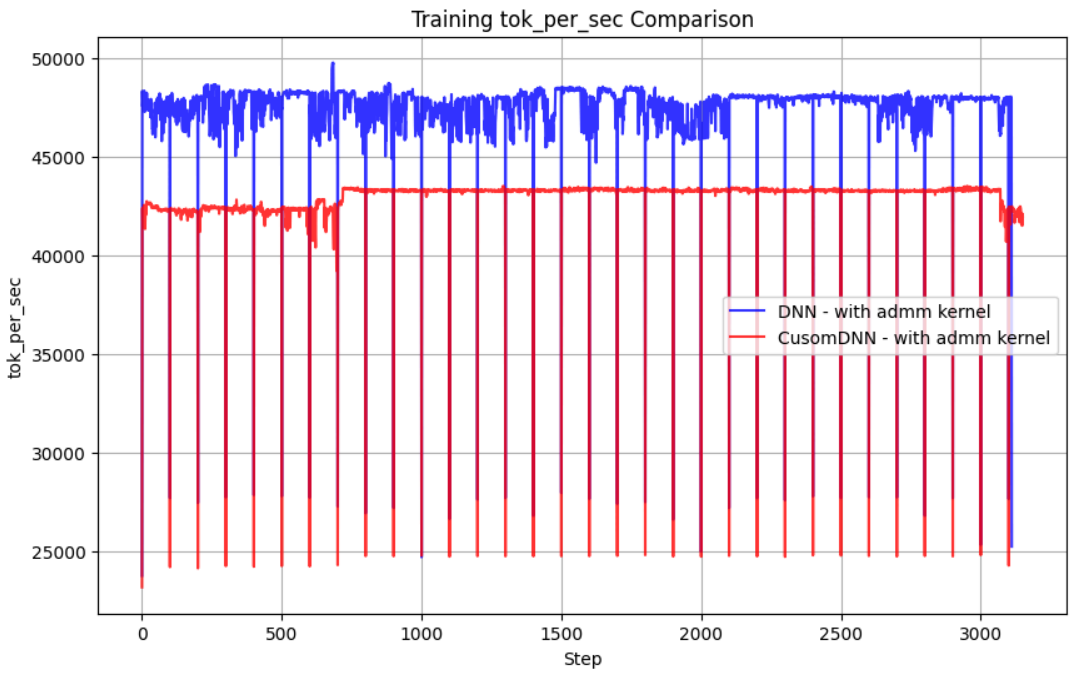

Getting a worse throughput then Adhitya - maybe I have few extra sync calls.

This could also be due to the continuous training is heating the GPUs, thus hampering the throughput.

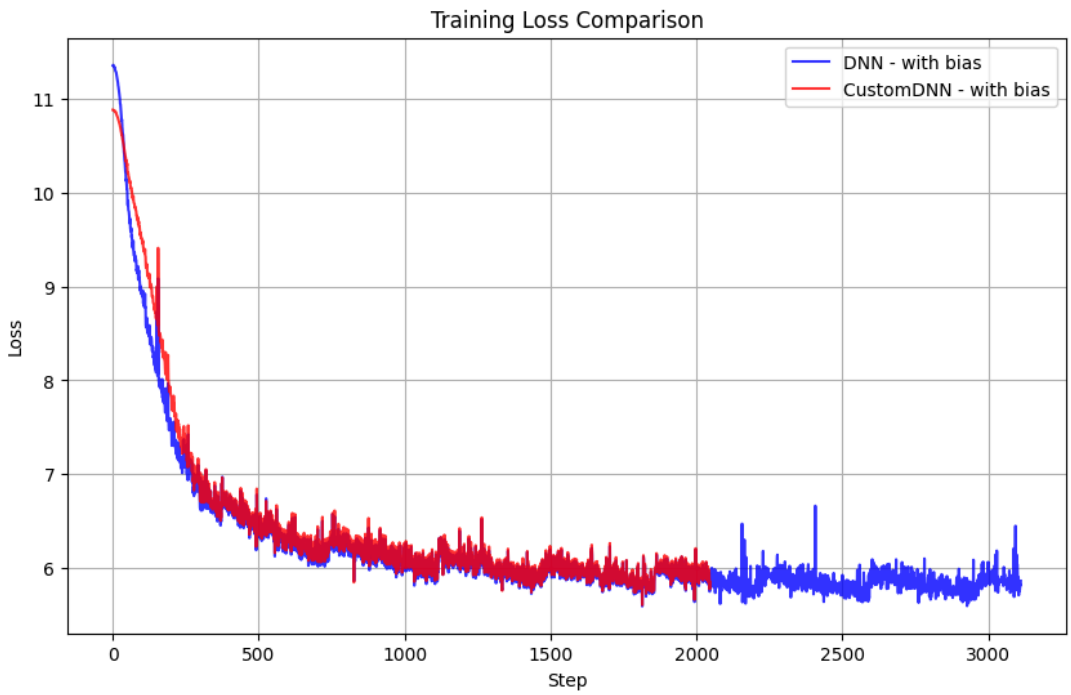
Although I got a more stable throughput curve.



Did the same after removing the addmm kernel and using the default kernels.

The training is still incomplete, but again following the same trend - deviation followed by overlap.

Again the throughput is lesser for me.



```
Maximum throughput (DNN - with bias):  49066.98779
Avg. Throughput (DNN - with bias): 46821.978077118256
Min Train Loss (DNN - with bias): 5.588697
Average loss (DNN - with bias):  6.245370461760927
Max Grad Norm (DNN - with bias):  3.782947
Min Val Loss (DNN - with bias): 5.841533

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Maximum throughput (CustomDNN - with bias):  44104.573475
Avg. Throughput (CustomDNN - with bias): 41927.807299461165
Min Train Loss (CustomDNN - with bias): 5.601429
Average loss (CustomDNN - with bias):  6.523812994137763
Max Grad Norm (CustomDNN - with bias):  2.906503
Min Val Loss (CustomDNN - with bias): 5.936807
```
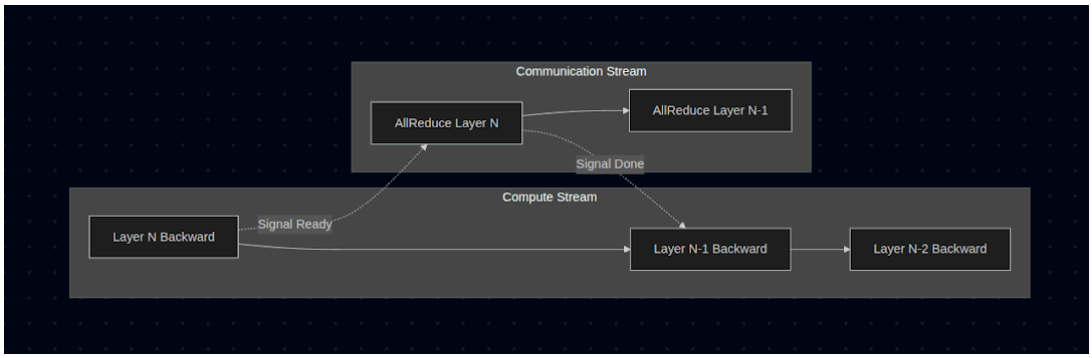

Multi-tasking on the GPU (Overlapping)

- Overlapping Communication with Computation is simply the act of having the GPU do math for the next step while the network handles sending data from the last step.
    - Compute Stream - Matrix Multiplications, or GEMMs
    - Communication Stream - sending data across GPUs (NCCL AllReduce).
- Because they are in separate lanes, they need a way to talk to each other so they don't get out of sync.
- They do this using "breadcrumbs" called CUDA Events.
- As seen in the diagram below, when the Compute Stream finishes calculating Layer N Backward, it drops Signal Ready
- The Communication Stream sees this breadcrumb and immediately starts sending that layer's data across the network (AllReduce Layer N).
- Once the network transfer is complete, it drops Signal Done so the Compute Stream knows it's safe to move on to the next layer (Layer N-1 Backward).



- In Tensor Parallelism, there is a "hard dependency" we need the final network result of Layer A before we can start the math for Layer B.
- To achieve this, we slice the layer
- This is called Granularity Splitting or Chunking.

- Instead of doing one massive math operation (a giant GEMM) and then one massive network transfer (AllReduce), we break that batch of data into smaller chunks -
    - Compute math for Chunk 1.
    - While the network is transmitting Chunk 1 across the GPUs, the compute cores immediately start doing the math for Chunk 2

Trade-offs -

a. SM Contention -

- Both chunks of work are fighting for the same Streaming Multiprocessors
- If network tasks take up too many workers, math tasks slow down.

b. Bandwidth Saturation

- No matter how cleverly you overlap, data cables can only carry a limited data per second (PCIe or NVLink Bandwidth limits).

c. CPU Overhead

- The CPU is the boss giving out these parallel tasks.
- If it assigns tasks too slowly, the GPU workers will still end up waiting around idle (known as a stream bubble)

Regards

Anuj Kumar